BELGIUM CAMPUS iTversity	**Project Charter - Template**	TAB

Programme: AR/VR	**Project Code:** 3VR
Project Title: Immersive AR/VR Campus Tour Mobile Application for Belgium Campus Open Day	**Supervisor:** TBA

1 Research Programme

Project Definition:

Belgium Campus hosts multiple Open Day events annually across its campuses (Pretoria, Kempton Park, Stellenbosch), offering prospective students and parents the opportunity to explore facilities, academic programmes, innovation labs, and student life. While physical tours are available, many attendees experience time constraints, overcrowding, or are unable to attend in person.



This project seeks to design and develop a mobile-based AR/VR immersive campus tour application that enables prospective students and parents to explore campus facilities virtually using smartphones and VR headsets available at the campus. The system will integrate 360° panoramic tours, interactive information hotspots (e.g., Robotics Lab, IoT Lab, Lecture Halls, Library, Student Accommodation), and optional audio narration to enhance engagement.

The application will support both:

- On-site Open Day VR stations (using campus VR headsets and APK installations)
- Remote access via Android mobile devices

The project will result in a functional immersive system that enhances the Open Day experience through digital innovation and experiential learning.

Research Goals:

- Immersive Engagement: Investigate how AR/VR technologies can improve prospective student engagement during university Open Day events.
- Interactive Information Delivery: Explore the effectiveness of interactive 360° environments with embedded digital hotspots for communicating institutional information.
- Technology Integration in Higher Education Marketing: Evaluate how immersive technologies can support recruitment, branding, and digital transformation strategies in higher education.

Methodology:

- **Literature Review**: Review existing research on immersive learning environments, AR/VR in higher education marketing, virtual campus tours, and user engagement theory.

- **Requirements Analysis**: Gather stakeholder requirements from Open Day coordinators, marketing teams, IT department, and prospective student personas.

- **System Design**: Design system architecture including:
  - Mobile AR/VR application architecture
  - 360° content management structure
  - Hotspot interaction logic
  - Backend analytics (optional)

- **Prototype Development**: Develop a functional APK-based Android application integrating:
  - 360° panoramic images/videos
  - Interactive hotspots
  - Audio narration
  - VR headset compatibility mode

- **Usability Testing**: Conduct usability testing during a simulated Open Day or pilot group to assess:
  - User engagement
  - Navigation ease
  - Information clarity
  - System performance

- **Iterative Improvement**: Refine the system based on user feedback and technical evaluation.

Project Objectives:

1. Develop a mobile-based AR/VR application capable of displaying immersive 360° campus environments.
2. Integrate interactive hotspots that provide multimedia information (text, images, video, audio narration) about key campus facilities.
3. Ensure compatibility with Android devices and campus VR headsets via APK deployment.
4. Design a scalable architecture that allows future addition of new campuses, facilities, or Open Day content.
5. Evaluate the system's effectiveness in improving engagement and information retention among prospective students.

Project Scope

In-Scope:

- Capture and integration of 360° images/videos of campus facilities.
- Development of Android-based AR/VR mobile application.
- Integration of interactive hotspots with multimedia overlays.
- VR headset compatibility testing.
- Deployment via APK installation.
- Usability evaluation with Open Day participants.
- Basic analytics (e.g., most viewed locations, interaction counts).

Out-of-Scope:

- Development of custom VR hardware.
- iOS version development (unless time permits).
  Full institutional marketing system redesign.
- Live real-time multi-user virtual environment (multiplayer VR).

Deliverables

1. **Functional AR/VR Mobile Application (APK):** Fully operational immersive campus tour app compatible with Android and VR headsets.
2. **360° Digital Campus Tour Content Library:** Structured repository of panoramic media assets and associated metadata.
3. **System Architecture & Technical Documentation:** Including UML diagrams, architecture diagrams, API documentation, and deployment guide.
4. **Usability Report:** Analysis of user engagement, feedback, performance metrics, and recommendations.
5. **Demonstration & Deployment Plan:** On-site VR setup plan for Open Day including installation instructions and maintenance guidelines.

Tools & Techniques

- **Development Platforms:**
  - Unity 3D (for VR/AR development)
  - Unreal Engine (alternative option)
  - Android Studio (for APK packaging)
- **AR/VR Frameworks:**
  - Google VR SDK
  - OpenXR
  - ARCore (if augmented functionality is added)
- **360° Content Creation:**
  - Insta360 or Ricoh Theta cameras
  - Adobe Premiere / After Effects (video editing)
  - Photoshop (image optimisation)
- **Backend & Analytics (Optional):**
  - Firebase
  - Node.js + Express
  - MongoDB or Firebase Firestore
- **UI/UX Design:**
  - Figma (prototyping)
  - User journey mapping
- **Testing & Evaluation:**
  - Usability testing protocols
  - SUS (System Usability Scale)
  - Performance benchmarking

References

https://www.getcampusview.com
https://360tourpro.com/universities-offering-360-virtual-tours/
https://campus360.org/ar/for-education-institutions
https://www.campusxr.org/

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